

YOURSTORY



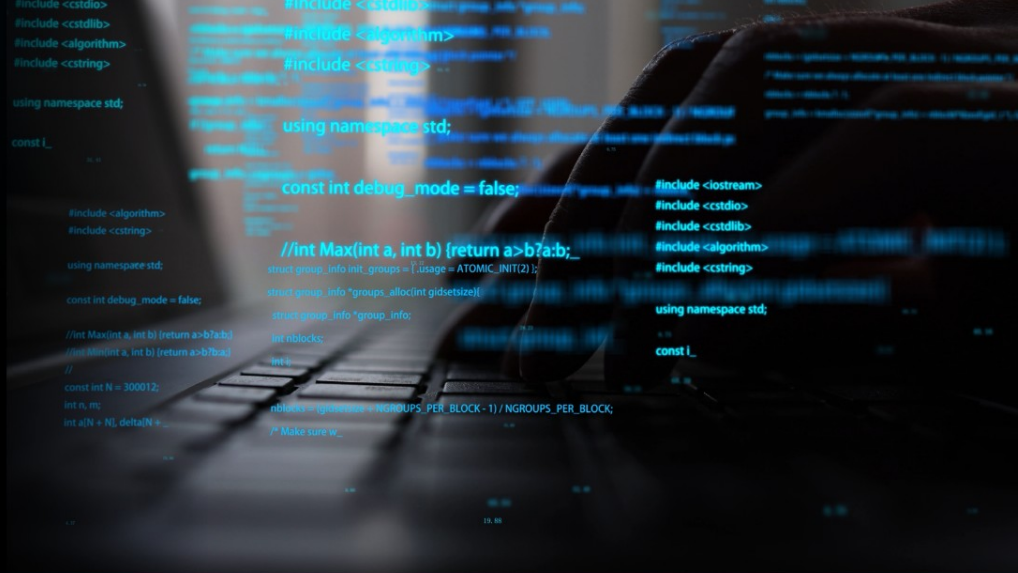
H2S



Coco CLI

HACKATHON

GCC Edition



Team Name : **Provenance**

Team Leader Name : **Manoj T**

Team Size : **2**

Problem Statement : **1 — Risk, Fraud and Regulatory Intelligence Copilot**

PaperTrail — the problem

Risk, Fraud and Regulatory Intelligence Copilot · banking and NBFC

Finding the risk signal is the cheap part. **Defending the number is the expensive part.**

THE PAIN TODAY

An analyst answers “which transactions breached our AML thresholds last month” in ten minutes — then spends two days assembling an evidence pack an auditor will accept.

Dashboards hand you numbers nobody can defend. Two analysts asked the same question compute it differently: booking or value date, reversals in or out.

THE USER

A compliance analyst or MLRO at a mid-size commercial bank — the person who has to put their name on the number.

Banking is India's largest GCC segment; regulatory reporting and AML operations are the day job of tens of thousands of those staff.

What PaperTrail produces is not a chat reply. It is a filing — and every figure in it is a footnote that resolves to its governed metric definition, the exact SQL, the source rows, and the regulation clause that makes it matter.

Architecture

Snowflake-native end to end · no LangChain, no external vector store, no third-party LLM API

STRUCTURED · synthetic banking data

240,798 transactions · 809 counterparties · 1,497 accounts · 18 months

UNSTRUCTURED · 11 regulatory PDFs

Parsed inside Snowflake with AI_PARSE_DOCUMENT → 72 clauses

RAW → CURATED → GOLD · 9 Dynamic Tables

Entity resolution; governance choices kept as explicit columns, not filtered away

CORTEX SEARCH · REGULATION_SEARCH

Clause retrieval with citation attributes · 11/12 top-1, 12/12 top-3

SEMANTIC VIEW · Cortex Analyst

7 governed metrics · 10 verified queries · each metric states its governance decision

ROW ACCESS POLICIES

AU / SG analyst personas, justified against AUSTRAC Pt 8.1 and MAS Notice 626 §13

CORTEX AGENT figures → Analyst · rules → Search · both when a number needs a rule · refuses what it cannot ground

risk-scanner

governed query +
provenance bundle

regulation-linker

clause retrieval
with citations

finding-writer

assembles finding;
cannot invent a figure

provenance-logger

append-only,
content-hashed

Four reusable CoCo skills, each with a documented interface and an explicit boundary. finding-writer cannot state an ungrounded figure: its prompt contains only extracted facts with no database access, and every number in its output is validated back to an input fact.

We measured the claim everyone else asserts

Governed vs ungoverned — same data, same model, full raw DDL, no hint of any governance rule

Our hypothesis was wrong. An ungoverned model is not inconsistent — across 25 runs the SQL was byte-identical. It is perfectly confident and materially wrong on 2 of 5 questions.

QUESTION	UNGOVERNED	GOVERNED	
Suspicious volume, Q3 2026	8,797,923	4,295,229	+105%
Exposure to CP-STRUCT-01	87,185	87,185	agree*
Alert closure rate 2026	63.6%	63.6%	agree
SAR filing share	6.5%	6.5%	agree
High-risk counterparties, Singapore	23	36	-36%

* **Q2 agrees by coincidence** — it computes direct exposure rather than entity-resolved, and matches only because that counterparty has no beneficial-ownership links. Agreement is not evidence of a sound method.

The \$4.5M overstatement comes from four missed decisions at once: booking date rather than value date, reversals counted, unsettled counted, and transactions on already-closed alerts treated as suspicious. Nothing in the output signals any of it.

Measured outcomes

Scored against an answer key that was never loaded into Snowflake

0.944

detection recall
17 of 18 planted positives

5 of 6

typologies at F1 1.000

11/12

clause retrieval, top-1
12/12 at top-3

5/5

agent tool-routing
including the refusal

TIME SAVED, HONESTLY STATED

The evidence pack a finding produces — figure, metric definition, SQL, source rows, clause — is the artefact that takes an analyst two days to assemble by hand. PaperTrail emits it with the answer, and the figure re-derives from its own footnote in about 300 ms.

WHY THIS IS NOT 100%

Velocity-spike detection is weak — 20 false positives. Removing it would take overall precision and F1 to 1.000. We kept it: velocity monitoring is a regulatory requirement, not an optional enhancement. A perfect score on self-authored data, detectors and answer key would be evidence of tuning, not detection.

Scalability and beyond the demo. Every object is idempotent DDL and every row is generated, so the system rebuilds byte-for-byte in any Snowflake account. The four skills are parameterised for reuse: swap the semantic view and the clause corpus, keep the provenance machinery.

Live app papertrail-provenance.vercel.app no login required · Repo github.com/tjmanoj/papertrail

Native app PAPERTRAIL.GOLD.PAPERTRAIL_APP · Streamlit in Snowflake

CoCo evidence docs/coco/ — 11 full session transcripts and 26 verbatim prompts, each re-derivable with cortex conversations transcript <id>

YOURSTORY



1125

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THANK YOU



```
#include <algorithm>
#include <string>

using namespace std;

const int debug_mode = false;

#include <algorithm>
#include <string>

using namespace std;

const int debug_mode = false;

//int Max(int a, int b) {return a>b?a:b;}
//int Min(int a, int b) {return a<b?a:b;}
//
const int N = 300012;
int n, m;
int a[N + N], delta[N + N];

// Make sure w
```